

- 4.4 A gas of 4 kg is contained within a piston-cylinder machine. The gas undergoes a process for which $pV^{1.5} = \text{constant}$. The initial pressure is 3 bar and the initial volume is 0.1 m^3 , and the final volume is 0.2 m^3 . The specific internal energy of the gas decreases by 4.6 kJ/kg . There are no significant changes in KE and PE. Determine the net heat transfer for the process.

Hint:

$$W_{1-2} = \frac{P_1 V_1 - P_2 V_2}{n - 1} \text{ where } P_2 = P_1 \left(\frac{V_1}{V_2} \right)^n = 3 \left(\frac{0.1}{0.2} \right)^{1.5} = 1.06 \text{ bar}$$

$$= +17.6 \text{ kJ}$$

$$Q = \Delta U + W = m(u_2 - u_1) + W = 4 \times (-4.6) + 17.6$$

$$= -0.8 \text{ kJ Ans.}$$

- 4.5 A piston-cylinder assembly contains 5 kg of steam. The steam having an internal energy of 2709.9 kJ/kg expands to a state where the internal energy is 2659.6 kJ/kg . During the process, there is heat transfer of 80 kJ to the steam and also a paddle-wheel work transfer of 18.5 kJ . Neglecting KE and PE of steam, determine the amount of energy transfer by work from the steam to the piston and the decrease in system energy.

Hint:

$$Q = \Delta U + W = \Delta U + W_{pw} + W_{pav}$$

$$\therefore W_{pdv} = Q - m(u_2 - u_1) - W_{pw} = 350 \text{ kJ Ans.}$$

$$\Delta E = \text{Energy input} - \text{Energy output} = -251.5 \text{ kJ Ans.}$$

- 4.6 The rate of heat transfer between a certain electric motor and its surroundings varies with time as $\dot{Q} = -0.2 [1 - e^{-0.65t}]$ where t is in seconds and $\dot{Q}$ is in kW. The shaft of the motor rotates at a constant speed of 100 rad/s and applies a constant torque of 18 Nm to an external load. The motor draws a constant electrical power input of 2 kW . Obtain an expression for the time rate of change of energy of the motor.

Hint: The time rate of change of system energy

$$\frac{dE}{dt} = \dot{Q} - \dot{W} = \dot{Q} - (\dot{W}_{\text{shaft}} + \dot{W}_{\text{elect}})$$

$$\dot{W}_{\text{shaft}} = Tw = 0.08 \times 100 \times 10^{-3} = +1.8 \text{ kW}$$

$$\dot{W}_{\text{elect}} = -2.0 \text{ kW}$$

$$\therefore \frac{dE}{dt} = 0.2 [1 - e^{-0.65t}] + 1.8 - 2$$

$$= 0.2 e^{-0.65t} \text{ Ans.}$$

PROBLEMS

- 4.1 An engine is tested by means of a water brake at 1000 rpm. The measured torque of the engine is 10000 mN and the water consumption of the brake is $0.5 \text{ m}^3/\text{s}$, its inlet temperature being 20°C . Calculate the water temperature at exit, assuming that the whole of the engine power is ultimately transformed into heat which is absorbed by the cooling water. (Ans. 20.5°C)
- 4.2 In a cyclic process, heat transfers are $+14.7 \text{ kJ}$, -25.2 kJ , -3.56 kJ and $+31.5 \text{ kJ}$. What is the net work for this cyclic process? (Ans. 17.34 kJ)
- 4.3 A slow chemical reaction takes place in a fluid at the constant pressure of 0.1 MPa . The fluid is surrounded by a perfect heat insulator during the reaction which begins at state 1 and ends at state 2. The insulation is then removed and 105 kJ of heat flow to the surroundings as the fluid goes to state 3. The following data are observed for the fluid at states 1, 2 and 3:

State	$V(\text{m}^3)$	$t(^{\circ}\text{C})$
1	0.003	20
2	0.3	370
3	0.06	20

For the fluid system, calculate E_2 and E_3 , if $E_1 = 0$

(Ans. $E_2 = -29.7 \text{ kJ}$, $E_3 = -110.7 \text{ kJ}$)

- 4.4 During one cycle the working fluid in an engine engages in two work interactions: 15 kJ to the fluid and 44 kJ from the fluid, and three heat interactions, two of which are known: 75 kJ to the fluid and 40 kJ from the fluid. Evaluate the magnitude and direction of the third heat transfer. (Ans. -6 kJ)
- 4.5 A domestic refrigerator is loaded with food and the door closed. During a certain period the machine consumes 1 kW h of energy and the internal energy of the system drops by 5000 kJ . Find the net heat transfer for the system. (Ans. -8.6 MJ)
- 4.6 1.5 kg of liquid having a constant specific heat of 2.5 kJ/kg K is stirred in a well-insulated chamber causing the temperature to rise by 15°C . Find ΔE and W for the process. (Ans. $\Delta E = 56.25 \text{ kJ}$, $W = -56.25 \text{ kJ}$)
- 4.7 The same liquid as in Problem 4.6 is stirred in a conducting chamber. During the process 1.7 kJ of heat are transferred from the liquid to the surroundings, while the temperature of the liquid is rising to 15°C . Find ΔE and W for the process. (Ans. $\Delta E = 54.55 \text{ kJ}$, $W = 56.25 \text{ kJ}$)
- 4.8 The properties of a certain fluid are related as follows:
- $$u = 196 + 0.718 t$$
- $$pv = 0.287 (t + 273)$$
- where u is the specific internal energy (kJ/kg), t is in $^\circ\text{C}$, p is pressure (kN/m^2), and v is specific volume (m^3/kg). For this fluid, find c_v and c_p . (Ans. 0.718 , 1.005 kJ/kg K)
- 4.9 A system composed of 2 kg of the above fluid expands in a frictionless piston and cylinder machine from an initial state of 1 MPa , 100°C to a final temperature of 30°C . If there is no heat transfer, find the net work for the process. (Ans. 100.52 kJ)
- 4.10 If all the work in the expansion of Problem 4.9 is done on the moving piston, show that the equation representing the path of the expansion in the pv -plane is given by $pv^{1.4} = \text{constant}$.
- 4.11 A stationary system consisting of 2 kg of the fluid of Problem 4.8 expands in an adiabatic process according to $pv^{1.2} = \text{constant}$. The initial conditions are 1 MPa and

200°C, and the final pressure is 0.1 MPa. Find W and ΔE for the process. Why is the work transfer not equal to $\int p dV$?

(Ans. $W = 217.35$, $\Delta E = -217.35$ kJ, $\int p dV = 434.4$ kJ)

- 4.12 A mixture of gases expands at constant pressure from 1 MPa, 0.03 m³ to 0.06 m³ with 84 kJ positive heat transfer. There is no work other than that done on a piston. Find ΔE for the gaseous mixture. (Ans. 54 kJ)

The same mixture expands through the same state path while a stirring device does 21 kJ of work on the system. Find ΔE , W , and Q for the process.

(Ans. 54 kJ, -21 kJ, 33 kJ)

- 4.13 A mass of 8 kg gas expands within a flexible container so that the p - v relationship is of the form $pv^{1.2} = \text{const}$. The initial pressure is 1000 kPa and the initial volume is 1 m³. The final pressure is 5 kPa. If specific internal energy of the gas decreases by 40 kJ/kg, find the heat transfer in magnitude and direction. (Ans. +2615 kJ)

- 4.14 A gas of mass 1.5 kg undergoes a quasi-static expansion which follows a relationship $p = a + bV$, where a and b are constants. The initial and final pressures are 1000 kPa and 200 kPa respectively and the corresponding volumes are 0.20 m³ and 1.20 m³. The specific internal energy of the gas is given by the relation,

$$u = 1.5 pv - 85 \text{ kJ/kg}$$

where p is in kPa and v is in m³/kg. Calculate the net heat transfer and the maximum internal energy of the gas attained during expansion. (Ans. 660 kJ, 503.3 kJ)

- 4.15 The heat capacity at constant pressure of a certain system is a function of temperature only and may be expressed as

$$C_p = 2.093 + \frac{41.87}{t + 100} \text{ J/}^\circ\text{C}$$

where t is the temperature of the system in °C. The system is heated while it is maintained at a pressure of 1 atmosphere until its volume increases from 2000 cm³ to 2400 cm³ and its temperature increases from 0°C to 100°C. (a) Find the magnitude of the heat interaction. (b) How much does the internal energy of the system increase? (Ans. (a) 238.32 J (b) 197.79 J)

- 4.16 An imaginary engine receives heat and does work on a slowly moving piston at such rates that the cycle of operation of 1 kg of working fluid can be represented as a circle 10 cm in diameter on a p - v diagram on which 1 cm = 300 kPa and 1 cm = 0.1 m³/kg. (a) How much work is done by each kg of working fluid for each cycle of operation? (b) The thermal efficiency of an engine is defined as the ratio of work done and heat input in a cycle. If the heat rejected by the engine in a cycle is 1000 kJ per kg of working fluid, what would be its thermal efficiency?

(Ans. (a) 2356.19 kJ/kg, (b) 0.702)

- 4.17 A gas undergoes a thermodynamic cycle consisting of three processes beginning at an initial state where $p_1 = 1$ bar, $V_1 = 1.5$ m³ and $U_1 = 512$ kJ. The processes are as follows:

(a) Process 1-2: Compression with $pV = \text{constant}$ to $p_2 = 2$ bar, $U_2 = 690$ kJ

(b) Process 2-3: $W_{23} = 0$, $Q_{23} = -150$ kJ, and

(c) Process 3-1: $W_{31} = +50$ kJ. Neglecting KE and PE changes, determine the heat interactions Q_{12} and Q_{31} .

- 4.18 A gas undergoes a thermodynamic cycle consisting of the following processes: (i) Process 1-2: Constant pressure $p = 1.4$ bar, $V_1 = 0.028$ m³, $W_{12} = 10.5$ kJ, (ii) Process 2-3: Compression with $pV = \text{constant}$, $U_3 = U_2$, (iii) Process 3-1: Constant volume, $U_1 - U_3 = -26.4$ kJ. There are no significant changes in KE and PE. (a) Sketch

(Ans. 74 kJ, 22 kJ)

the cycle on a p - V diagram. (b) Calculate the net work for the cycle in kJ. (c) Calculate the heat transfer for process 1-2 (d) Show that $\sum_{\text{cycle}} Q = \sum_{\text{cycle}} W$.

(Ans. (b) - 8.28 kJ, (c) 36.9 kJ)

- 4.19 A fluid contained in a cylinder receives 150 kJ of mechanical energy by means of a paddle wheel, together with 50 kJ in the form of heat. At the same time, a piston in the cylinder moves in such a way that the pressure remains constant at 200 kN/m² during the fluid expansion from 2m³ to 5m³. What is the change in internal energy and in enthalpy? (Ans. - 400 kJ, + 200 kJ)